Adaptive Spatial Reasoning with Structured Visual Tokens

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Abstract

Generative visual models can follow broad textual descriptions while still missing exact spatial constraints. We study a structured representation that separates object identity from position, scale, and orientation, then use it to guide generation and revision. The method is designed to make spatial requirements inspectable before rendering and to support targeted corrections without regenerating the entire scene. This sample document demonstrates the report layout; all names, claims, and numerical values are placeholders and should be replaced with verified project material.

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1 Introduction

Text-to-image systems have improved substantially, but precise spatial control remains difficult. Large pretrained models provide a powerful basis for visual generation (Radford et al., 2021), yet a prompt can name the right objects and still produce the wrong arrangement, scale, or relative position. This gap limits use cases where the visual output must satisfy a layout or scene specification.

We represent a scene as a set of object records. Each record contains a label and a normalized bounding box, so users can inspect and edit spatial intent before image generation. The representation is deliberately simple: it can be created from language, edited directly, and translated into constraints for a downstream generator.

2 Method

For an image of width W and height H, an object i is represented by $o_i = (c_i, x_i, y_i, w_i, h_i)$, where c_i is its category and the remaining values are normalized coordinates. The center of the object is $(x_i W, y_i H)$ and its size is $(w_i W, h_i H)$. A scene is the collection $\mathcal{S} = \{o_i\}_{i=1}^{N}$.

An instruction such as "move the cup to the left of the plate" can be encoded as an ordering constraint:

$$x_{\text{cup}} + \frac{w_{\text{cup}}}{2} < x_{\text{plate}} - \frac{w_{\text{plate}}}{2}. \tag{1}$$

The layout can then be checked against explicit constraints before it is sent to an image generator.

2.1 A small example

Table 1 shows example object records. The values are illustrative and are included only to demonstrate table styling.

Table 1: Example normalized object layout.

Object	x	y	w	h
Cup	0.28	0.62	0.12	0.20
Plate	0.66	0.62	0.26	0.18

3 Evaluation

A useful evaluation should measure both visual quality and constraint satisfaction. For example, a benchmark can report the fraction of generated scenes that satisfy all specified relations, alongside human preference or a perceptual quality metric. Reporting only an aggregate image score can obscure whether the system followed the requested layout.

4 Limitations

Bounding boxes do not fully describe occlusion, articulated pose, or complex three-dimensional relations. Some instructions are also ambiguous without additional context. A practical system should allow users to inspect the parsed layout, clarify uncertain constraints, and revise individual objects.

5 Conclusion

Structured scene records provide a compact interface between language and visual generation. They make spatial requirements easier to inspect and can support targeted editing. The example content in this template should be replaced with the report's own methods, evidence, and limitations.

References

Alec Radford et al. Learning transferable visual models from natural language supervision. In *Proceedings of the International Conference on Machine Learning*, pages 8748–8763, 2021.

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